

Riley Hill

Electrical Engineering · Protection & Control · Renewables Integration

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scan → portfolio

Electrical Engineering senior specializing in power-system protection & control and renewables integration. FE (Electrical & Computer) certified. Portfolio of hardware-validated protective relays, coordination and arc-flash studies, SCADA systems, and solar hosting-capacity analysis.

EDUCATION

Texas A&M University — B.S. Electrical Engineering, College Station, TX *Aug 2023 – May 2027*

- **FE Exam (Electrical & Computer) — Passed April 2026**
- Relevant coursework: Power System Analysis (ECEN 459), Senior Design Capstone (ECEN 403)

TECHNICAL SKILLS

Languages & Tools: Python, MATLAB, Simulink, C/C++, Verilog, pandapower, NumPy/matplotlib

Protection & Power: IEC 60255, IEC 60909, IEEE 1584, NFPA 70E, IEEE 1547, ANSI C84.1, symmetrical components, load flow, short-circuit, device coordination, arc-flash

SCADA & Comms: Ignition (Perspective), Modbus TCP, DNP3, OPC tags

Hardware & Design: Oscilloscope, soldering, current sensing, closed-loop control; KiCad, Altium, SolidWorks, AutoCAD, LTspice

PROJECTS

Full interactive portfolio: rileyhill625.github.io

IDMT Overcurrent Relay (Hardware) — Designed and built a microprocessor-based inverse-time overcurrent relay implementing IEC 60255 trip curves in firmware; validated trip timing on hardware (192 mA → 1.1 s) and commissioned it with a purpose-built closed-loop current-injection test set (~1% accuracy).

Distribution Feeder Protection Study — Performed short-circuit, protective-device coordination, and IEEE 1584 arc-flash studies on a 12.47 kV feeder in Python/pandapower, producing TCC coordination curves and NFPA 70E PPE categories.

SCADA Monitoring System — Built relay-to-control-room telemetry: a Modbus TCP server feeding an Ignition Perspective HMI over OPC with overcurrent alarming and event logging, plus a Modbus/DNP3 protocol comparison study.

Solar-plus-Storage Feeder Study — Modeled solar PV hosting capacity and a battery-storage dispatch controller on a distribution feeder in pandapower; flattened the net-load duck curve, eliminated midday over-voltage, and shaved evening peak demand ~42%.

EXPERIENCE

Electronics Technician | Texas A&M Chemistry Department, College Station, TX *[Mon. Year] – Present*

- Diagnosed, repaired, and rebuilt laboratory instrumentation; installed a replacement relay and performed board-level soldering and rework.
- Reverse-engineered a perfboard circuit into a manufacturable PCB in KiCad.

Video Engineer | 12th Man Productions, College Station, TX *Oct 2023 – Sept 2025*

- Engineered and operated live broadcast video systems for NCAA Division I athletic events, delivering broadcast-quality output under live conditions.
- Configured switchers and camera shading/color; collaborated with producers, directors, and engineers across the production team.

Delivery Driver | Amazon DSP, College Station, TX *Summer 2026*

- Completed high-volume delivery routes on tight schedules with a consistent safety and reliability record.

LEADERSHIP & INVOLVEMENT

Treasurer, Theta Tau (professional engineering fraternity), 1 year · Volunteer: The Big Event, Habitat for Humanity, Aggieland Humane Society